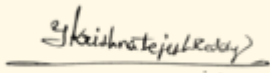


SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO2 ASSESSMENT TOOL 2 – Architecture Design Assignment

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO2	Assessment:	Assessment Tool 2 – Architecture Design Assignment
Weightage:	20%	Total Marks:	25
CO2 Statement:	<i>Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills</i>		
Student Name:	Y.Krishna Tejesh Reddy	Reg. No.:	192411200
Date:	3-8-2026	Duration:	60 minutes

Code of Conduct

I Y.Krishna Tejesh Reddy(192411200) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the Student: _____

Annexure A – Architecture Design Assignment

1. Analyze System Requirements

The **Campus Placement and Recruitment Management Portal** is a web-based application developed to simplify and automate the campus placement process. It provides a common platform where students, recruiters, placement officers, and administrators can efficiently manage placement activities. The system replaces manual processes with an online solution, improving communication, reducing paperwork, and increasing transparency.

- **System Objectives**

- Automate the campus placement process.
- Maintain student and recruiter information digitally.
- Enable online job posting and job applications.
- Simplify interview scheduling and notifications.
- Generate placement reports.
- Improve efficiency and transparency.

❖ **Functional Requirements**

- Student registration and login.
- Resume upload and profile management.
- Job posting by recruiters.
- Online job application.
- Interview scheduling.
- Application status tracking.
- Notification management.
- Report generation. .

○ **Non-Functional Requirements**

- Fast system response.
- Secure authentication and data protection.
- Reliable system with regular backups.
- User-friendly interface.
- Scalable architecture for future growth.
- Easy maintenance and updates.
- High system availability.

❖ **Quality Attributes**

The proposed system should provide **scalability** to support more users, **security** to protect confidential information, **performance** for quick response time, **reliability** for uninterrupted services, **maintainability** for easy software updates, and **availability** to ensure users can access the portal whenever required..

2. Select a Suitable Software Architecture

For the **Campus Placement and Recruitment Management Portal**, the most suitable software architecture is the **Layered Architecture (Three-Tier Architecture)**. This architecture divides the system into separate layers, namely the Presentation Layer, Business Logic Layer, and Database Layer. Each layer performs a specific function, making the system well-organized, secure, and easy to maintain.

The **Presentation Layer** provides the user interface where students, recruiters, placement officers, and administrators interact with the system. It allows users to register, log in, apply for jobs, post vacancies, and view placement information.

The **Business Logic Layer** processes user requests and implements the core functionality of the system. It validates user information, checks job eligibility, manages applications, schedules interviews, generates reports, and sends notifications.

The **Database Layer** stores all important information such as student profiles, recruiter details, job postings, applications, interview schedules, and placement records. It ensures secure storage and quick retrieval of data whenever required.

The Layered Architecture is suitable because it separates the user interface, business logic, and database operations, making the application easier to develop, test, and maintain. It also provides better security by restricting direct access to the database and improves system performance through efficient data management.

In the future, additional features such as online aptitude tests, AI-based resume screening, mobile application support, and video interview modules can be integrated without affecting the existing system. Therefore, the Layered Architecture is the most appropriate choice for developing a secure, scalable, reliable, and maintainable Campus Placement and Recruitment Management Portal.

3. Draw the Software Architecture Diagram

Software Architecture Overview

Main Components

- **Presentation Layer (User Interface)**
 - Student Portal
 - Recruiter Portal
 - Placement Officer Portal
 - Admin Portal
- **Business Logic Layer**
 - User Authentication
 - Job Management
 - Student Management
 - Application Management

- Interview Scheduling
- Notification Service
- Report Generation
- **Database Layer**
 - Student Database
 - Recruiter Database
 - Job Database
 - Placement Database
 - Interview Database
- **External Services**
 - Email Notification Service
 - SMS Notification Service

Component Interaction

- Students log in and apply for jobs through the Student Portal.
- Recruiters post job openings and shortlist candidates.
- Placement Officers verify student details and manage placement drives.
- The Business Logic Layer processes all requests and validates data.
- The Database Layer stores and retrieves all information.
- Email and SMS services send notifications to users.

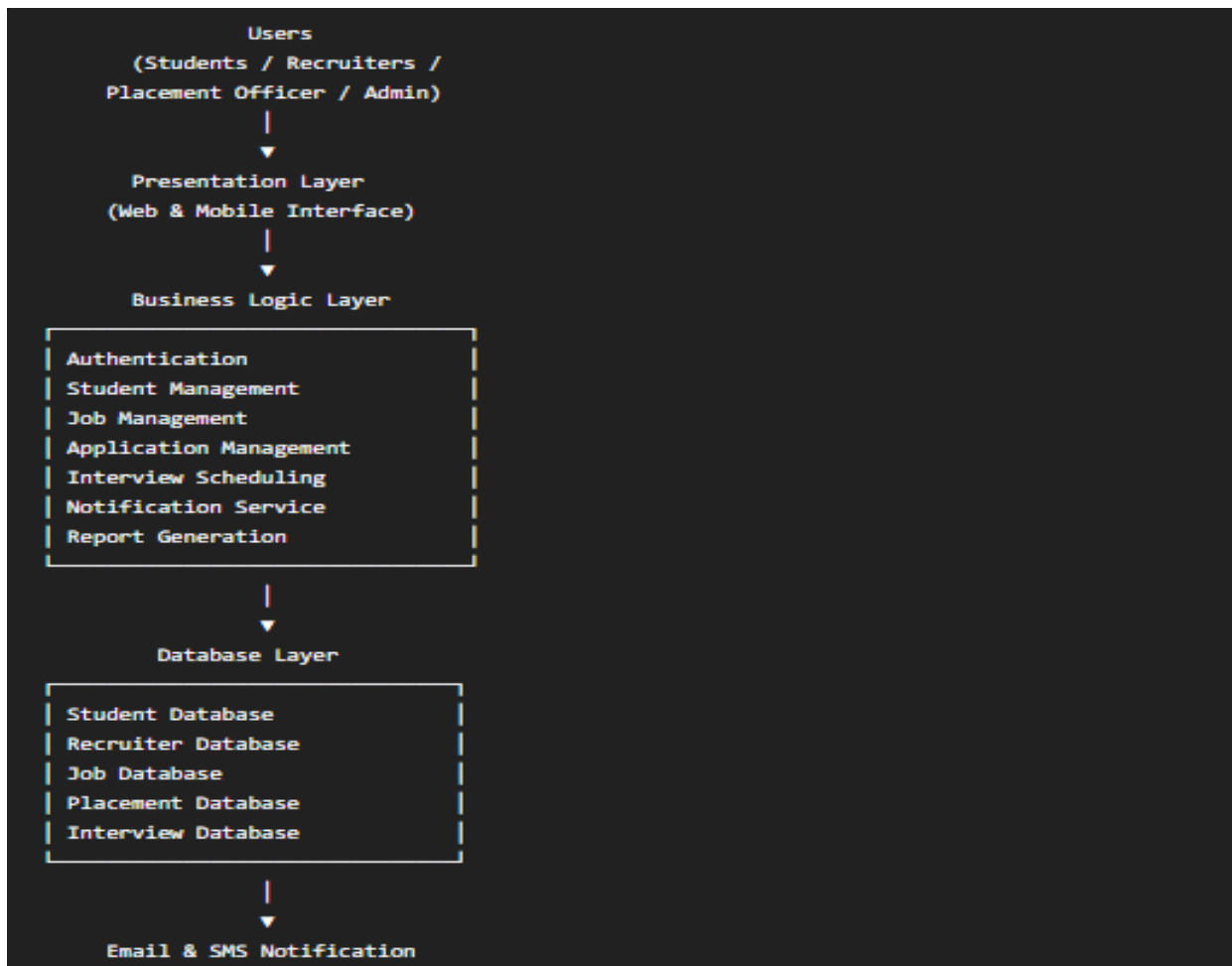
System Workflow

1. Student registers and logs into the portal.
2. Recruiter posts available job openings.
3. Student searches and applies for eligible jobs.
4. Placement Officer verifies applications.
5. Recruiter shortlists candidates.
6. Interview schedules are generated.
7. Students receive interview notifications.
8. Final selection results are published.
9. Placement reports are generated and stored.

Benefits of the Architecture

- Easy communication between all modules.
- Secure data management.
- Faster processing of user requests.
- Simple maintenance and upgrades.

- Supports future enhancements.
- Improves overall system performance.



4. Explain Components and Their Interactions

The Campus Placement and Recruitment Management Portal consists of several components that work together to automate the placement process. Each component has a specific responsibility and communicates with other components to ensure smooth system operation.

Components of the System

1. Student Module

- Student registration and login.
- Profile creation and update.
- Resume upload.

- Search and apply for jobs.
- Track application status.
- View interview schedules.

2. Recruiter Module

- Company registration.
- Post job vacancies.
- View eligible student profiles.
- Shortlist candidates.
- Schedule interviews.
- Publish selection results.

3. Placement Officer Module

- Verify student profiles.
- Approve recruiter registrations.
- Manage placement drives.
- Monitor recruitment activities.
- Generate placement reports.
- Send notifications.

4. Administrator Module

- Manage users and roles.
- Monitor system performance.
- Manage database.
- Backup and restore data.
- Configure system settings.

5. Database

- Stores student information.
- Stores recruiter details.
- Maintains job postings.
- Stores application records.
- Maintains interview schedules.
- Generates placement reports.

Component Interaction

- Students register and log in to access placement services.
- Recruiters post job openings through the recruiter portal.
- Students apply for eligible jobs.
- Placement Officers verify applications and manage placement activities.
- The Business Logic Layer processes all requests and updates the database.
- The Database stores and retrieves information whenever required.
- Email and SMS services send notifications regarding interviews and selection results.

Overall System Workflow

1. Student creates an account and uploads the resume.
2. Recruiter registers and posts job opportunities.
3. Students apply for suitable jobs.
4. Placement Officer verifies applications.
5. Recruiter shortlists eligible candidates.
6. Interview schedules are prepared.
7. Notifications are sent to students.
8. Interview is conducted.
9. Final results are published.
10. Placement reports are generated.

Benefits of Component Interaction

- Ensures smooth communication between all users.
- Reduces manual work and paperwork.
- Improves placement process efficiency.
- Provides secure data management.
- Enables quick report generation.
- Supports future system enhancements.

5. Discuss Quality Attributes

The proposed **Layered Architecture** satisfies the important quality attributes required for the Campus Placement and Recruitment Management Portal. These attributes ensure that the system performs efficiently, remains secure, and supports future enhancements.

Scalability

- Supports an increasing number of students and recruiters.
- Can handle multiple placement drives simultaneously.
- Allows new modules to be added easily.
- Suitable for future expansion.

Security

- Provides secure user authentication.
- Uses role-based access control.
- Protects confidential student and recruiter data.
- Prevents unauthorized access to the system.

Performance

- Provides fast response to user requests.
- Retrieves and stores data efficiently.
- Handles multiple users with minimum delay.
- Optimizes database operations.

Reliability

- Ensures accurate processing of placement activities.
- Maintains data consistency.
- Supports automatic backup and recovery.
- Reduces system failures and downtime.

Maintainability

- Modular architecture makes updates easier.
- Simplifies bug fixing and testing.

- Allows developers to modify individual modules without affecting the entire system.
- Reduces maintenance effort.

Availability

- Provides 24×7 access to users.
- Supports continuous placement activities.
- Ensures high system uptime.
- Allows users to access the portal from anywhere with an internet connection.

6. Justify Architectural Decisions.

The **Layered (Three-Tier) Architecture** was selected because it provides a clear separation between the user interface, business logic, and database. This architecture improves the overall organization of the system and makes development, testing, and maintenance easier.

Reasons for Selecting the Architecture

- Separates the system into independent layers.
- Simplifies software development and maintenance.
- Improves system security.
- Supports efficient data management.
- Enhances code reusability.
- Provides better performance.
- Makes debugging and testing easier.
- Supports future system expansion.

Trade-offs Considered

- Development may require more time due to multiple layers.
 - Communication between layers may introduce slight processing overhead.
 - Initial system design is more complex than a simple application.
 - However, the long-term benefits outweigh these limitations.
-

Support for Future Enhancements

The selected architecture allows new features to be added without affecting the existing system.

Future enhancements include:

- AI-based Resume Screening.
- Online Aptitude Test Module.
- Video Interview Integration.
- Mobile Application Support.
- Chatbot for Student Assistance.
- Placement Analytics Dashboard.
- Cloud-Based Deployment.
- Third-Party API Integration.

System Integration

The architecture supports integration with external systems such as:

- Email Notification Service.
- SMS Gateway.
- College ERP System.
- Student Information System.
- Cloud Database Services.

Long-Term Maintainability

- Easy to update individual modules.
- Simplifies software maintenance.
- Reduces development cost.
- Improves software reliability.
- Enables quick implementation of future requirements.
- Ensures smooth operation for a large number of users.

Conclusion

The **Layered (Three-Tier) Architecture** is the most suitable choice for the Campus Placement and Recruitment Management Portal because it provides **security, scalability, maintainability, reliability, performance, and flexibility**. It effectively supports the current system requirements while allowing future enhancements and integration with other technologies. This architecture ensures that the portal remains efficient, secure, and easy to maintain throughout its lifecycle.